5-1. Regression - brief revision

- We study the relationship between a **response** (or **dependent**) random variable and one or more **explanatory** (or **independent**) variables.
- The value assumed by the response variable Y is, in part, explained by the values of the other variables, $X_1, X_2, \ldots$
  For example:
  - Height of mother (x) and height of daughter (Y)
  - Selling price of a house (Y) may depend upon total floor area (x_1), number of bedrooms (x_2), age of the house (x_3), etc.
- We are only concerned here with the theory of *linear* relationships.
- Data: n pairs (x_i, y_i) , $i = 1, 2, \ldots, n$ [*Simple* Linear Regression]
- Basic display is a *scatterplot* – this may exhibit pattern, indicative of an underlying relationship between the variables.

It is convenient to define the **summary statistics**

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2, \qquad S_{yy} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

[Note: for calculations,

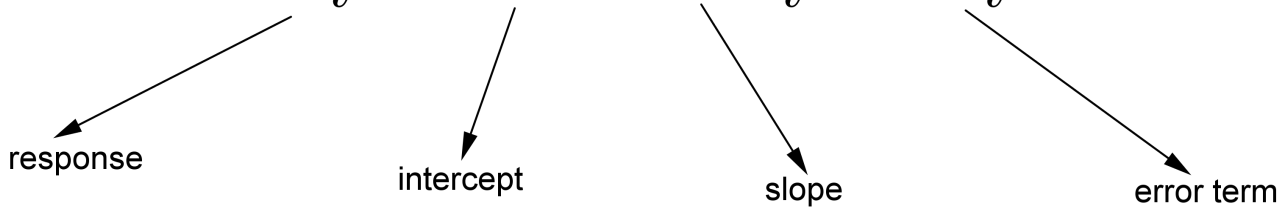
$$S_{xx} = \sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2, \qquad S_{yy} = \sum y_i^2 - \frac{1}{n}\left(\sum y_i\right)^2,$$

$$S_{xy} = \sum x_i y_i - \frac{1}{n}\left(\sum x_i\right)\left(\sum y_i\right)\Big]$$

Linear regression analysis

The basic linear model with a single explanatory variable is:

$$Y_i = a + bx_i + \varepsilon_i$$



a, b are non-random parameters.

x_i, the explanatory variable values, are non-random.

ε_i is a random variable, the "error term", a "residual" random variable with mean 0 and variance σ^2. The ε_i's are *uncorrelated*.

The response variable values Y_i are random because of the randomness in the ε_i values.

- Another way to express the model is:

$$E[Y_i|x_i] = a + bx_i$$

  with uncorrelated errors with constant variance σ^2

- It follows that $V[Y_i] = \sigma^2$ and the Y_i's are uncorrelated.
- 'Homoscedastic' means that the Y_i all have the same variance, σ^2

Estimation of parameters

Use *Least Squares* – i.e. minimise

$$S = \sum_{i=1}^{n} (y_i - a - bx_i)^2$$

to obtain estimators

$$\boxed{\hat{b} = \frac{S_{xy}}{S_{xx}} \quad \textbf{and} \quad \hat{a} = \bar{y} - \hat{b}\bar{x}}$$

The **fitted value** at $X = x$ is:

$$\hat{y} = \hat{a} + \hat{b}x$$

and we note that the fitted line goes through $(\bar{x}, \bar{y})$.

The **residual** *from the fit* for observation (x_i, y_i) is the estimated error:

$$e_i = \hat{\varepsilon}_i = y_i - \hat{y}_i$$

Notice that

$$\begin{aligned} \epsilon_i &= y_i - (a + bx_i) \\ e_i &= y_i - \left(\hat{a} + \hat{b}x_i\right) \end{aligned}$$

The sum of squared residuals is given as

$$\sum_i e_i^2 = \sum (y_i - \hat{y}_i)^2 = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$$

Properties of Parameter estimates

$$E\left[\hat{b}\right] = b, \quad V\left[\hat{b}\right] = \frac{1}{S_{xx}}\sigma^2$$

$$E\left[\hat{a}\right] = a, \quad V\left[\hat{a}\right] = \left(\frac{\sum x_i^2}{nS_{xx}}\right)\sigma^2$$

(Notice that estimates are *unbiased*.)

$$Cov\left[\hat{a}, \hat{b}\right] = -\left(\frac{\sum x_i}{nS_{xx}}\right)\sigma^2$$

Note: $Cov = 0$ if $\sum x_i = 0$ – and this is easily achieved (centre x values by defining a new variable $x_i' = x_i - \bar{x}$, and notice $\sum x_i' = \sum (x_i - \bar{x}) = 0$).

Also:

$$\widehat{\sigma^2} = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n-2}\left(S_{yy} - \frac{S_{xy}^2}{S_{xx}}\right)$$

and

$$E\left[\widehat{\sigma^2}\right] = \sigma^2 \qquad \text{(unbiased)}$$

Partitioning the total variability in the responses

$$\begin{aligned} \sum(y_i - \bar{y})^2 &= \sum(y_i - \hat{y}_i)^2 + \sum(\hat{y}_i - \bar{y})^2 \\ SS_T &= SS_{RES} + SS_{REG} \end{aligned}$$

- SS_T : "total sum of squares" – total variability in the responses
- SS_{RES} : "residual (or error) sum of squares" – variability remaining "unexplained" by the model
- SS_{REG}: "regression sum of squares" – variability "explained" by the regression model

Coefficient of determination

The proportion of the total variability in the responses "explained" by a model is called the *coefficient of determination*, R^2:

$$\boxed{R^2 = \frac{SS_{REG}}{SS_T}} = \frac{S_{xy}^2}{S_{xx}S_{xy}}$$

For Simple Linear Regression, $R^2 = r^2$, the sample correlation between x and y

Inference on parameters: full normal model

- Now further assume: $\varepsilon_i \sim N(0, \sigma^2)$ for all i
  i.e. the errors are i.i.d. *normal* random variables.
- Then the response r.v.s Y_i are independent with

$$Y_i \sim N(a + bx_i, \sigma^2)$$

- We can then use Maximum Likelihood (ML) estimation: estimates of a and b same as with Least Squares (LS), but

$$MLE\left(\sigma^2\right) = \frac{\sum e_i^2}{n} = \frac{n-2}{n}\widehat{\sigma^2} \qquad (biased)$$

Inference on parameters

- Under the *normal model* we have:

$$\hat{b} \sim N\left(b, \frac{\sigma^2}{S_{xx}}\right) \quad \text{giving} \quad \frac{\hat{b} - b}{(\sigma^2/S_{xx})^{1/2}} \sim N(0,1)$$

- With unknown variance:

$$\frac{\hat{b} - b}{\left(\widehat{\sigma^2}/S_{xx}\right)^{1/2}} \sim t_{n-2}$$

- Similarly for parameter a. *Exercise*: check details, using estimated standard error of $\hat{a}$ given earlier.

- Also $\hat{a}$, $\hat{b}$ independent of $\widehat{\sigma^2}$ and

$$\frac{(n-2)\widehat{\sigma^2}}{\sigma^2} \sim \chi^2_{n-2}$$

Inference on responses

- **For mean response:**

  At x_0, the *mean (expected)* response μ_0 is estimated as

$$\hat{\mu}_0 = \hat{E}(Y|x_0) = \hat{a} + \hat{b}x_0$$

  and under the *normal model*, we have

$$\frac{\hat{\mu}_0 - \mu_0}{e.s.e.(\hat{\mu}_0)} \sim t_{n-2}$$

  where

$$e.s.e.\,(\hat{\mu}_0) = \left[\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\widehat{\sigma^2}\right]^{1/2}$$

- **For individual response:**

  At x_0, the *individual* response y_0 is estimated as before

$$\hat{y}_0 = \hat{a} + \hat{b}x_0$$

  but *uncertainty* about this estimate is now *greater*.

  Have

$$\frac{\hat{y}_0 - y_0}{e.s.e.\,(\hat{y}_0)} \sim t_{n-2}$$

  with

$$e.s.e. = \left[\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\widehat{\sigma^2}\right]^{1/2}$$